TUIA • The Universal Isotope Atlas

[illegible]

[illegible]: [illegible]98] = [illegible]UPAC 2021 [illegible] = [illegible]

[illegible] = [illegible]NIST / IUPAC 2021 / Wikipedia + NUBASE2020 [illegible] tools/scoring.py

hydrogen S95
H
1
[illegible]H
1 99.99%
2 0.0115%
1.008
[1.00784, 1.00811]
[illegible]

helium S88
He
2
4 100%
3 <0.01%
4.0026
[illegible]

lithium A70
Li
3
7 92.41%
6 7.59%
6.94
[6.938, 6.997]
[illegible]

beryllium A77
Be
4
[illegible]Be
9 100%
9.0122

boron A70
B
5
11 80.1%
10 19.9%
10.81
[10.806, 10.821]
[illegible]

carbon B56
C
6
[illegible]
12 98.93%
13 1.07%
12.011
[12.0096, 12.0116]
[illegible]

nitrogen S95
N
7
14 99.64%
15 0.364%
14.007
[14.00643, 14.00728]
[illegible]

oxygen S95
O
8
16 99.76%
18 0.205%
17 0.038%
15.999
[15.99903, 15.99977]
[illegible]

fluorine S95
F
9
19 100%
18.9984

neon B53
Ne
10
20 90.48%
22 9.25%
21 0.27%
20.1797
[illegible]

sodium S95
Na
11
23 100%
22.9898

magnesium S88
Mg
12
24 78.99%
26 11.01%
25 10%
24.305
[24.304, 24.307]
[illegible]

aluminium S95
Al
13
[illegible]Al
27 100%
26.9815

silicon B55
Si
14
28 92.22%
30 3.092%
29 4.685%
28.085
[28.084, 28.086]
[illegible]

phosphorus S95
P
15
31 100%
30.9738

sulfur S88
S
16
32 94.99%
36 0.01%
34 4.25%
33 0.75%
32.06
[32.059, 32.076]
[illegible]

chlorine A73
Cl
17
[illegible]Cl
35 75.76%
37 24.24%
35.45
[35.446, 35.457]
[illegible]

argon A70
Ar
18
40 99.6%
36 0.3336%
38 0.0629%
39.95
[39.792, 39.963]
[illegible]

potassium S88
K
19
39 93.26%
41 6.73%
40 0.0117%
39.0983
[illegible]

calcium S88
Ca
20
40 96.94%
48 0.187%
46 <0.01%
44 2.086%
43 0.135%
42 0.647%
40.078
[illegible]

scandium A74
Sc
21
45 100%
44.9559

titanium A73
Ti
22
48 73.72%
46 8.25%
47 7.44%
50 5.18%
49 5.41%
47.867
[illegible]

vanadium A77
V
23
51 99.75%
50 0.25%
50.9415

chromium A73
Cr
24
52 83.79%
54 2.365%
50 4.345%
53 9.501%
51.9961
[illegible]

manganese S88
Mn
25
55 100%
54.938

iron A69
Fe
26
56 91.75%
58 0.282%
57 2.119%
54 5.845%
55.845
[illegible]

cobalt S88
Co
27
59 100%
58.9332

nickel A68
Ni
28
58 68.08%
64 0.9255%
62 3.635%
61 1.14%
60 26.22%
58.6934
[illegible]

copper A68
Cu
29
63 69.15%
65 30.85%
63.546
[illegible]

zinc B59
Zn
30
64 49.17%
70 0.61%
68 18.45%
67 4.04%
66 27.73%
65.38
[illegible]

gallium C46
Ga
31
69 60.11%
71 39.89%
69.723
[illegible]

germanium A68
Ge
32
70 20.57%
72 27.45%
76 7.73%
74 36.5%
73 7.75%
72.63
[illegible]

arsenic S88
As
33
75 100%
74.9216

selenium A69
Se
34
74 0.89%
76 9.37%
77 7.63%
78 23.77%
82 8.73%
80 49.61%
78.971
[illegible]

bromine A70
Br
35
79 50.69%
81 49.31%
79.904
[79.901, 79.907]
[illegible]

krypton A77
Kr
36
84 56.99%
78 0.355%
80 2.286%
82 11.59%
83 11.5%
86 17.28%
83.798
[illegible]

rubidium B51
Rb
37
85 72.17%
87 27.83%
85.4678
[illegible]

strontium B63
Sr
38
88 82.58%
84 0.56%
86 9.86%
87 7%
87.62
[illegible]

yttrium A77
Y
39
89 100%
88.9058

zirconium A68
Zr
40
90 51.45%
96 2.8%
94 17.38%
92 17.15%
91 11.22%
91.224
[illegible]

niobium A77
Nb
41
93 100%
92.9064

molybdenum A68
Mo
42
92 14.53%
94 9.15%
95 15.84%
100 9.82%
98 24.39%
97 9.6%
96 16.67%
95.95
[illegible]

technetium C36
Tc
43
98
[98]

ruthenium A70
Ru
44
96 5.54%
98 1.87%
99 12.76%
100 12.6%
101 17.06%
104 18.62%
102 31.55%
101.07
[illegible]

rhodium B61
Rh
45
103 100%
102.9055

palladium B59
Pd
46
102 1.02%
104 11.14%
105 22.33%
110 11.72%
108 26.46%
106 27.33%
106.42
[illegible]

silver C44
Ag
47
107 51.84%
109 48.16%
107.8682
[illegible]

cadmium B59
Cd
48
106 1.25%
108 0.89%
110 12.49%
111 12.8%
112 24.13%
116 7.49%
114 28.73%
113 12.22%
112.414
[illegible]

indium C44
In
49
115 95.71%
113 4.29%
114.818
[illegible]

tin A68
Sn
50
120 32.58%
112 0.97%
114 0.66%
115 0.34%
116 14.54%
117 7.68%
118 24.22%
124 5.79%
122 4.63%
119 8.59%
118.71
[illegible]

antimony B50
Sb
51
121 57.21%
123 42.79%
121.76
[illegible]

tellurium A68
Te
52
120 0.09%
122 2.55%
123 0.89%
124 4.74%
125 7.07%
126 18.84%
130 34.08%
128 31.74%
127.6
[illegible]

iodine S95
I
53
[illegible]I
127 100%
126.9045

xenon A70
Xe
54
124 0.0952%
126 0.089%
128 1.91%
129 26.4%
130 4.071%
131 21.23%
136 8.857%
134 10.44%
132 26.91%
131.293
[illegible]

caesium S88
Cs
55
133 100%
132.9055

barium B63
Ba
56
138 71.7%
130 0.106%
132 0.101%
134 2.417%
135 6.592%
136 7.854%
137 11.23%
137.327
[illegible]

hafnium A68
Hf
72
[illegible]
174 0.16%
176 5.26%
177 18.6%
178 27.28%
179 13.62%
180 35.08%
178.49
[illegible]

tantalum A77
Ta
73
181 99.99%
180 0.01201%
180.9479
[illegible]

tungsten A68
W
74
[illegible]
180 0.12%
182 26.5%
183 14.31%
184 30.64%
186 28.43%
183.84
[illegible]

rhenium C35
Re
75
187 62.6%
185 37.4%
186.207
[illegible]

osmium B59
Os
76
[illegible]
184 0.02%
186 1.59%
187 1.96%
188 13.24%
189 16.15%
190 26.26%
192 40.78%
190.23
[illegible]

iridium C38
Ir
77
193 62.7%
191 37.3%
192.217
[illegible]

platinum B59
Pt
78
[illegible]
190 0.012%
192 0.782%
194 32.86%
198 7.356%
196 25.21%
195 33.78%
195.084
[illegible]

gold S88
Au
79
197 100%
196.9666

mercury C50
Hg
80
[illegible]
196 0.15%
198 9.97%
199 16.87%
200 23.1%
204 6.87%
202 29.86%
201 13.18%
200.592
[illegible]

thallium C35
Tl
81
205 70.48%
203 29.52%
204.38
[204.382, 204.385]
[illegible]

lead A76
Pb
82
208 52.4%
204 1.4%
206 24.1%
207 22.1%
207.2
[illegible]

bismuth S88
Bi
83
209 100%
208.9804

polonium D24
Po
84
209
[209]

astatine E15
At
85
210
[210]

radon D21
Rn
86
222
[222]

francium E15
Fr
87
223
[223]

radium D26
Ra
88
226
[226]

rutherfordium E11
Rf
104
267
—

dubnium E11
Db
105
268
—

seaborgium E11
Sg
106
269
—

bohrium E11
Bh
107
278
—

hassium E11
Hs
108
271
—

meitnerium E11
Mt
109
282
—

darmstadtium E11
Ds
110
282
—

roentgenium E11
Rg
111
286
—

copernicium E11
Cn
112
285
—

nihonium E11
Nh
113
286
—

flerovium E11
Fl
114
290
—

moscovium E11
Mc
115
290
—

livermorium E11
Lv
116
293
—

tennessine E11
Ts
117
294
—

oganesson E11
Og
118
294
—

[illegible]/ [illegible]7–71[illegible]89–103[illegible]

lanthanum A77
La
57
139 99.91%
138 0.08881%
138.9055
[illegible]

cerium A77
Ce
58
140 88.45%
136 0.185%
138 0.251%
142 11.11%
140.116
[illegible]

praseodymium A77
Pr
59
141 100%
140.9077

neodymium B59
Nd
60
[illegible]
142 27.15%
143 12.17%
150 5.638%
146 5.756%
148 17.19%
145 8.293%
144 23.8%
144.242
[illegible]

promethium D24
Pm
61
145
[145]

samarium B59
Sm
62
[illegible]
144 3.07%
147 14.99%
148 11.24%
149 13.82%
150 7.38%
154 22.75%
152 26.75%
150.36
[illegible]

europium B50
Eu
63
153 52.19%
151 47.81%
151.964
[illegible]

gadolinium B59
Gd
64
[illegible]
152 0.2%
154 2.18%
155 14.8%
156 20.47%
157 15.65%
160 21.86%
158 24.84%
157.25
[illegible]

terbium A77
Tb
65
159 100%
158.9254

dysprosium B59
Dy
66
[illegible]
156 0.056%
158 0.095%
160 2.329%
161 18.89%
162 25.47%
164 28.26%
163 24.9%
162.5
[illegible]

holmium A77
Ho
67
165 100%
164.9303

erbium B59
Er
68
[illegible]
162 0.139%
164 1.601%
166 33.5%
167 22.87%
170 14.91%
168 26.98%
167.259
[illegible]

thulium A77
Tm
69
169 100%
168.9342

ytterbium C50
Yb
70
[illegible]
168 0.123%
170 2.982%
171 14.09%
172 21.68%
173 16.1%
176 13%
174 32.03%
173.054
[illegible]

lutetium B51
Lu
71
175 97.4%
176 2.599%
174.9668
[illegible]

actinium D20
Ac
89
227
[227]

thorium A67
Th
90
232 100%
232.0377

protactinium D34
Pa
91
231 100%
231.0359

uranium B57
U
92
238 99.27%
234 <0.01%
235 0.7204%
238.0289
[illegible]

neptunium D28
Np
93
237
[237]

plutonium D29
Pu
94
244
[244]

americium D29
Am
95
243
—

curium D24
Cm
96
247
—

berkelium E18
Bk
97
247
—

californium E19
Cf
98
251
—

einsteinium E19
Es
99
252
—

fermium E15
Fm
100
257
—

mendelevium E15
Md
101
258
—

nobelium E15
No
102
259
—

lawrencium E15
Lr
103
266
—

[illegible]

[illegible] [illegible]nth [illegible]nth [illegible]nth [illegible]nth [illegible]nth [illegible] [illegible] [illegible]

[illegible] S [illegible] A [illegible] B [illegible] C [illegible] D [illegible] E [illegible] S [illegible] S [illegible]

[illegible]: [illegible] [illegible]: [illegible]

[illegible]: 98 [illegible]: [illegible]PmUNp [illegible]: [illegible]98] [illegible]

[illegible] 2 98.93% [illegible] 13 1.07%[illegible]56[illegible]W[illegible]—— [illegible] 2 [illegible] 99.9%[illegible]

[illegible]

[illegible]UIA [illegible]NIST / IUPAC 2021 [illegible]IUPAC IPTE[illegible]IUPAC [illegible] IUPAC [illegible]